



A model fusion method based DAT-DenseNet for classification and diagnosis of aortic dissection

Linlong He¹ · Shuhuan Wang¹ · Ruibo Liu¹ · Tienan Zhou² · He Ma^{1,3} · Xiaozeng Wang²

Received: 12 March 2024 / Accepted: 15 July 2024 / Published online: 5 September 2024
© Australasian College of Physical Scientists and Engineers in Medicine 2024

Abstract

In this paper, we proposed a complete study method to achieve accurate aortic dissection diagnosis at the patient level. Based on the CT angiography (CTA) images, a classification model named DAT-DenseNet, which combined the deep attention Transformer module with the DenseNet architecture is proposed. In the first phase, two DAT-DenseNet are combined in parallel. It is used to accurately achieve two classification task at the CTA images. In the second stage, we propose a feature fusion module. It concatenates and fuses the image features output from the two classification models on a patient by patient basis. In the comparison experiments of classification model performance, DAT-DenseNet obtained 92.41% accuracy at the image level, which was 2.20% higher than the commonly used model. In the comparison experiments of model fusion method, our method obtained 90.83% accuracy at the patient level. The experiments showed that DAT-DenseNet model exhibits high performance at the image level. Our feature fusion module achieves the mapping from two classification image features to patient outcomes. It achieves accurate patient classification. The experiments' results in the Discussion section elaborate the details of the experiment and confirmed that the results were reliable.

Keywords Aortic dissection · CT Angiography · Model fusion · Attention mechanism · Image classification

Introduction

Aortic dissection is an extremely dangerous cardiovascular disease [1]. In clinical, aortic dissection is defined as separation of the layers of the aortic wall due to the aorta or bleeding from the aortic wall [2]. According to statistical data, the mortality rate of patients with acute Stanford A aortic dissection is more than 50% within two days after the onset

of the disease, and more than 70% within one week after the onset of the disease. For acute Stanford B aortic dissection, the mortality rate can also reach 6.4% within 14 days after the onset of the disease. Unquestionably, aortic dissection is a serious threat to human life and health [3].

The rapid onset and high mortality rate of aortic dissection makes early diagnosis of aortic dissection particularly important. Today, the most common strategy for patients with suspected aortic dissection is computed tomography angiography (CTA). It is a laborious task for physicians

Linlong He and Shuhuan Wang have contributed equally to this work.

✉ He Ma
mahe@bmie.neu.edu.cn

✉ Xiaozeng Wang
wxiaozeng@163.com

Linlong He
h1766365160@163.com

Shuhuan Wang
2310523@stu.neu.edu.cn

Ruibo Liu
1910445@stu.neu.edu.cn

Tienan Zhou
zhoutienan810706@163.com

¹ College of Medicine and Biological Information Engineering, Northeastern University, Wenhua Road, Shenyang 110169, Liaoning, China

² State Key Laboratory of Frigid Zone Cardiovascular Diseases, Cardiovascular Research Institute and Department of Cardiology, General Hospital of Northern Theater Command, Wenhua Road, Shenyang 110016, Liaoning, China

³ The Key Laboratory of Intelligent Computing in Medical Image, Ministry of Education, Northeastern University, Wenhua Road, Shenyang 110819, Liaoning, China

to observe scanned images and then make a diagnosis [4]. With the rapid development of deep learning technology, the research on intelligent diagnosis of aortic dissection based on CTA images to be developed deeply. Now, the detection research on CT or CTA images for aortic dissection mainly takes the method of image transformation processing and deep learning. Lee et al. [5] proposed a method which combines multi-scale wavelet analysis and generative-discriminative model matching. It achieves segmentation of the true and false lumen of the aorta on CT images. Gayhart et al. [6] segmented the aorta based on the HU degree value and aortic's shape information on CTA images. Dehghan et al. [7] first segmented cross-sectional images of organs. Then operations such as segmentation refinement and flap detection were used to achieve aortic dissection classification at the slice level. Finally they mapped the slicing results to the patient level. In the study using deep learning methods, Yi et al. [8] utilized the U-Net model to achieve segmentation task, followed by scoring using the ResNet [9]. Finally, the Gaussian naive bayes model was used to achieve the discrimination between aortic and non-aortic dissection. Xu et al. [10] used Mask R-CNN model to segment the aortic contour. Edge detection technique was used to extract the aortic edges. Finally the task of classification of Stanford A aortic dissection was implemented using ResNet. Zhou et al. [11] proposed a deep model named MOLS-Net. It used ResNet in combination with the Sequential Feature Pyramid Attention Module to achieve fast and automatic segmentation of aortic dissection from CT images, followed by diagnosis using a multi-label classifier. Most of the above studies used segmentation models before diagnosis models. This requires a lot of data labelling work and reduces the diagnostic efficiency at the same time.

In medical research, multi-model fusion methods are used to predict and diagnose complex clinical diseases. Shen et al. [12] achieved coronary artery segmentation of CTA images by using three 3D fully convolutional networks (FCN) [13]. Akinori Hata et al. [14] proposed a model fusion method based Xception. The 2D slices were categorized into three categories: with aorta and containing aortic dissection, with aorta but not containing aortic dissection, and without aorta. Finally the patient categorization results were obtained using the maximum connectivity approach. Ding et al. [15] proposed a two-stage classification scheme based on ResNet and combined it with support vector machine (SVM) [16] to realize the classification task of leukocyte subtype. Vatsala et al. [17] proposed a fusion model fusing U-Net model and CNN model to segment the diseases in skin images and used CNN model to classify the segmented images into seven different skin disease categories. Wei et al. [18] proposed a model fusion method using DenseNet [19] and ConvNeXt as backbone subclassification models, and the features output from the sub-models were directly connected for skin

disease classification. Tomoki et al. [20] implemented locating the aorta and examining aortic lesions with two VGG [21] networks and achieved model fusion by setting a threshold for setting the probability score on the images to achieve result aggregation. Although the above studies used multiple models for disease diagnosis, no effective model fusion method was proposed, and the improvement in the diagnostic effectiveness of the disease was limited.

In order to solve the above problems, we propose a model fusion method for diagnosing patients with aortic dissection. The method is based on our CTA image dataset. The algorithm is divided into three steps. First, the original dataset is processed using a low-threshold zero-processing method. Second, two DAT-DenseNet classification models are trained using the processed image data: with or without aorta classification and with or without aortic dissection classification. Image features are extracted using the trained models. Then, in the model fusion session, the image features are spliced at the patient level and fed into the feature fusion module to achieve deep fusion of the output features of the two classification models, and finally the classification of aortic dissection patients is achieved. In summary, the contributions of this paper are as follows:

- DAT-DenseNet is proposed. It combines DenseNet and transformer multi-head attention module. It improves the problem of too large amount of parameters of traditional transformer module. Through the fusion of two DAT-DenseNet models, outstanding aortic dissection classification results are achieved without the need for aortic segmentation.
- A feature fusion module is proposed to achieve mapping of image features to patient features and deep fusion of features output from two classification models for patients, achieving excellent classification results at the patient level.

Methodology

The framework of the proposed algorithm for aortic dissection image classification based on CTA images is shown in Fig 1. The whole algorithm is divided into three main parts: data processing, classification model and feature fusion. In this section, we describe our algorithm by respectively illustrating the architecture of DAT-DenseNet, the structure of the DAT module and the feature fusion module.

DAT-DenseNet

The main idea of the DenseNet model is to establish a dense connection between all the layers in front and the layers behind, and to achieve feature reuse through feature splicing

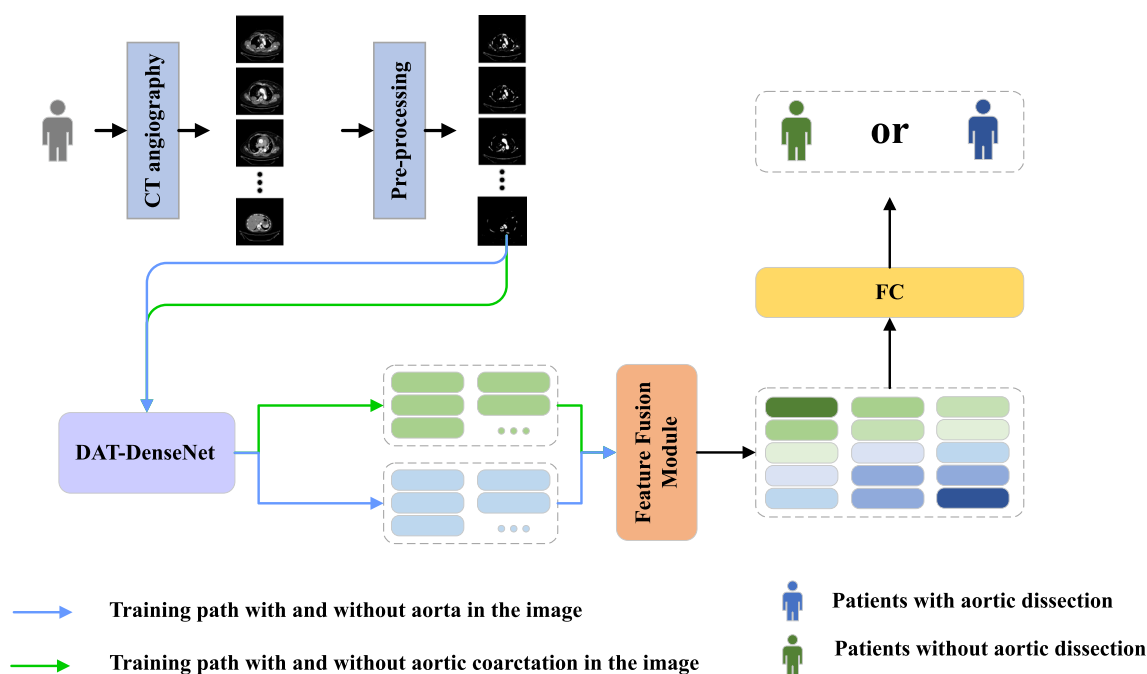


Fig. 1 Architecture of diagnostic algorithm for patients with aortic dissection

on the channel. This structure enhances the efficiency of utilising features in the network. However, for medical images, the location of lesion is usually fixed, using a dense convolutional neural network on medical images alone can easily lead to the model missing the contextual information of the lesions, thus affecting the performance of the model. To address the above problems, we improve DenseNet and proposed DAT-DenseNet, the overall architecture of DAT-DenseNet is shown in Fig 2. In the DAT-DenseNet model, the first convolutional layer, the batch normalization layer, the ReLU function and the maximum pooling layer in the first layer of DenseNet121 are replaced with the depth attention transformer module(DAT). In the DAT-DenseNet, the backbone is used as DenseNet-121 and the number of compression layers of the four-layer Dense Block are 6, 12, 24 and 16. The CTA image input to the model first passes through the DAT module. DAT relies on the Transformer architecture to achieve a larger sensory field for better

extraction of deep feature information and global information, and feature combination for flexibly generated patches. The extracted features are fed into the subsequent DenseNet network. The Dense Block module achieves high-level feature extraction by densely connecting multiple small-scale convolutions. The Transition module reduces the number of input feature map channels while fusing the features of each channel, and then compresses the size of the feature map using average pooling with a step size of 2. This combination enables the effective utilization of multi-level features to improve the classification accuracy.

DAT module

The structure of the DAT module is shown in Fig 3. The DAT module contains a convolutional embedding module, a depth convolution attention module (DCAM) and a residual united module (RUM). Convolutional embedding replaces

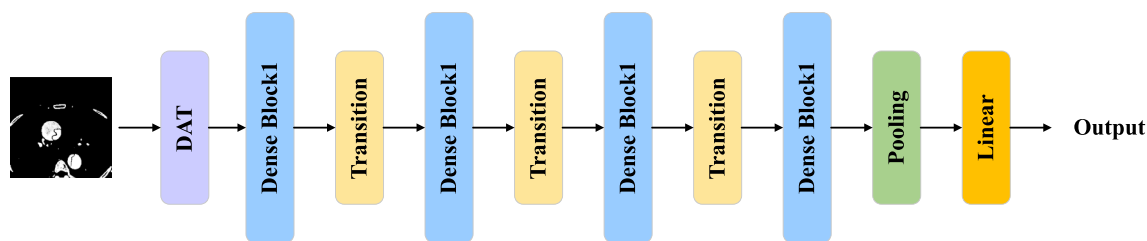


Fig. 2 Overall framework of the DAT-DenseNet

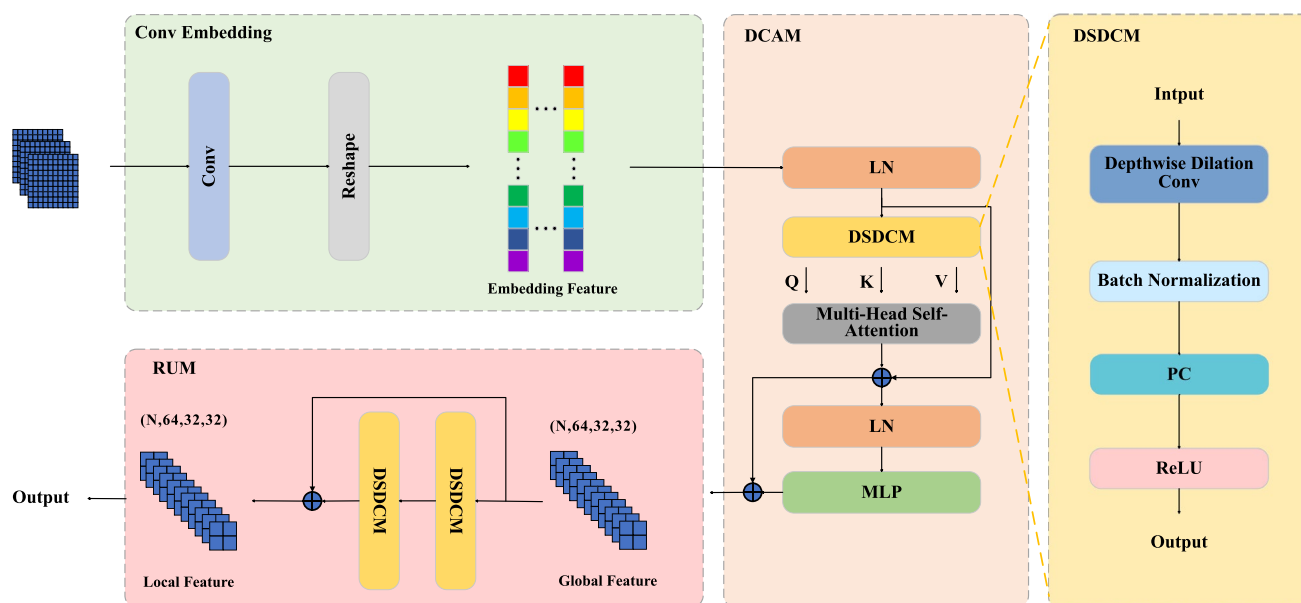


Fig. 3 Overall framework of the DAT module

the linear mapping in the standard transformer. It is responsible for permuting the input image into a number of patches, obtaining the local features by dilated convolution, and obtaining the mapped features after reshape operation. This operation greatly reduces the number of parameters in our model. The DCAM module consists of two layer normalization operations, a depthwise separable dilated convolution module (DSDCM), a multi-head attention module and a multilayer perceptron (MLP). In the DCAM, the convolution module used in DSDCM is a depth-separable null convolution module, which expands the sensory field. The input to the transformer's multi-head attention module is the feature vectors named q , k , and v . The multi-head attention mechanism can obtain more regional information of the image and realize a large enough sensory field. The obtained features are then fed into the MLP to learn the global information and perform residual jump concatenation with the input feature map. At the end of the DAT module, we introduce the residual unit module (RUM) to fuse features of different depths. This operation will effectively alleviate the problem of gradient vanishing during network training. Based on the above structure, the DAT module, as an attention module combining the Transformer architecture and null convolution, is able to achieve excellent feature extraction capability.

Feature fusion module

In the first stage of the algorithm, we trained two classification models using DAT-DenseNet based to achieve the classification of CTA images with and without aortic dissection, and with and without aorta. In the clinic, doctors cannot rely on

the results of one CTA image to judge the patient's condition for the diagnosis of medical images. Therefore, how to achieve effective aggregation of classification results from multiple images into the patient's classification results in the second stage is a major difficulty in the implementation of the model fusion method. To address this problem, this paper proposes a feature fusion module, which fuses the features extracted from two parallel classification models on a patient-by-patient basis, and finally aggregates them into a patient classification result. The specific structure of the fusion module is shown in Fig. 4. The fusion module mainly consists of a multi-head attention module and a MLP module. Two instance normalization (IN) transformations and two residual connections are set between them respectively. In the feature fusion module, the spliced features containing the outputs of the two classification models are first normalised. Then a Transformer module with multi-head self-attention is taken to implement the fusion of different modal features. The implementation of our extracted features in one of the sets of self-attention is illustrated in Fig. 4. IN considers all the elements of a single sample, a single channel, in calculating the normalised statistics to avoid neglecting the differences between different channels. MLP is used to finalise the classification of the feature parameters obtained from the self-attention module, and to compute the probability that the patient is an aortic dissection.

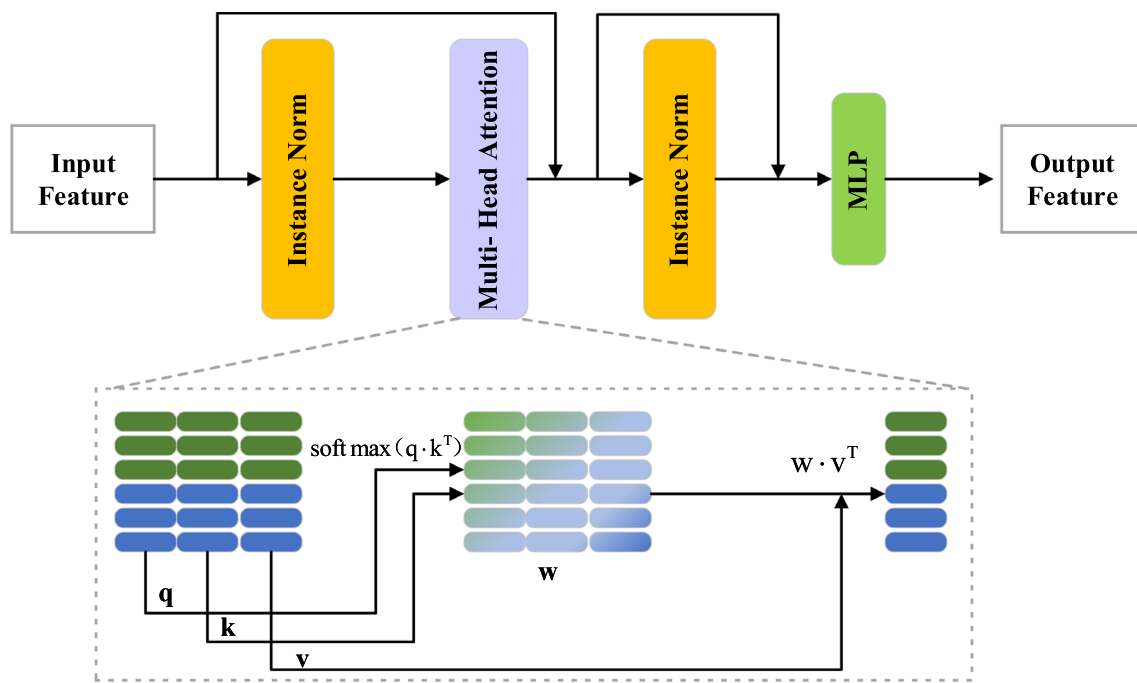


Fig. 4 Overall framework of the feature fusion module

Experiments and results

Datasets

In our experiments, the data of CTA images used contained two types of patients: 40 patients without aortic dissection and 40 patients with aortic dissection. After processing, a total of 98,166 2D images from 80 patients were obtained and 75,904 of these contained aortic dissection. These images were obtained from the results of CTA examinations of the aorta performed in 80 patients before the aortic surgery between 2015 and 2021. All patients were between 18 and 80 years of age. Patients with symptoms of abdominal wall thrombosis were excluded. The thickness of the image layer was 5 mm and the 2D images were all 512*512. All images were preprocessed using a low threshold zero processing method where the threshold was experimentally set to 200. The experiment was performed using a 5-fold cross-validation method. We randomly divide the patients into training set, validation set and

test set in the ratio of 6:2:2 to get the segmentation results of the images. The results of data allocation are shown in Table 1.

Evaluation metrics

To evaluate the performance of our method, several metrics are used to compare the classification results: Recall (REC), Precision (PRE), Specificity (SPE), F1-score (F1), accuracy (ACC). The above metrics are defined by True Negative (TN), False Positive (FP), False Negative (FN), and True Positive (TP) in the Confusion Matrix. Evaluation metrics are defined as follows:

$$Recall = \frac{TP}{TP + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

Table 1 Summary of Experimental Data on CTA Images

Type	Train	Valid	Test	Total
Number of patients with aortic dissection	24	8	8	40
Number of patients without aortic dissection	24	8	8	40
Number of patches with aortic dissection	13,357	4452	4453	22,262
Number of patches without aortic dissection	45,542	15,181	15,181	75,904

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (3)$$

$$F1 = 2 \times \frac{\text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} \quad (4)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

Implementation configurations

Our architecture is implemented on PyTorch. Most of the training experiments are performed on workstations equipped with Inter(R)Core(TM) i7-8700K CPU @ 3.70Ghz and NVIDIA GeForce RTX 1080 Ti GPU. A few comparison experiments are performed on workstations equipped with Inter(R)Core(TM) i7-13700KF CPU @ 3.40Ghz and NVIDIA GeForce RTX 4080 GPU. In the classification model, the AdamW [22] optimizer was used for training. The 5-fold cross-validation method was performed during training. The loss function used is the Cross-Entropy Loss.

Comparative experiments on classification models

In the comparison experiments, we set up two types of comparison experiments. One category is to compare with the classical CNN models, including AlexNet [23], VGG16, VGG19, ResNet18, ResNet50, ResNet101, DenseNet121, DenseNet161, DenseNet169. The other category is compared with the commonly used visual Transformer classification model, which contains ViT-B/16, ViT-B/32, Swin Transformer-B [24], and CvT [25]. The experimental results are shown in Table 2. These are the average of the evaluation metrics of the two classification models for the

with or without aorta and with or without aortic dissection. These DAT-DenseNet achieved an excellent performance of 85.75% precision, 82.66% F1 score and 92.41% accuracy at the slice level, and obtained the highest AUC index. At the same time, the model parameters of the DAT-DenseNet is limited to 7 M, which is lighter than most models. However, DAT-DenseNet slightly underperformance in recall and specificity.

Experiments to explore model fusion methods

We try to compare our approach with methods that use other model fusion. Our model fusion method combines two DAT-DenseNet models into a parallel architecture, where model output features are spliced on a patient-basis, and finally the feature fusion module is used for feature fusion and final diagnostic prediction. The experiment was still performed using five-fold cross-validation method. The results of the experiment are shown in Table 3, the values in table are patient-level results. The F1 reached 90.15%. Our model fusion method also yielded excellent classification results at the patient level, the ACC reached 90.83%. It can be seen that the our method obtained the best results for both F1 and ACC at the patient level. Compared to the other methods,

Table 3 Comparison of the performance of the model fusion methods

Method	REC (%)	PRE (%)	SPE (%)	F1 (%)	ACC (%)
Ding's	80.63	77.69	85.64	78.02	85.33
Wei's	81.96	75.37	86.19	81.03	86.24
Vatsala's	87.84	83.10	89.48	84.71	87.01
Akinori's	85.90	89.54	92.67	86.69	87.52
Our	89.62	91.03	91.37	90.15	90.83

Table 2 Performance analysis of the DAT-DenseNet along with the traditional models on test set

Model	REC (%)	PRE (%)	SPE (%)	F1 (%)	ACC (%)	Parameters(M)
AlexNet	61.88	73.74	88.05	66.90	84.84	57
VGG16	85.42	70.19	90.56	77.06	89.50	134
VGG19	63.55	80.75	96.32	71.12	89.91	140
ResNet18	49.40	71.59	84.94	56.42	81.77	11
ResNet50	70.32	82.53	94.69	75.94	88.29	24
ResNet101	63.55	79.15	95.93	70.50	89.60	42
DenseNet121	85.77	69.70	91.25	76.90	90.21	7
DenseNet161	44.60	77.98	94.35	56.78	78.96	26
DenseNet169	44.18	79.94	95.03	56.91	79.29	12
ViT-B/16	68.41	59.61	89.92	63.70	57.16	57
ViT-B/32	38.39	40.12	90.33	39.24	82.82	59
Swin Transformer-B	63.72	47.99	88.34	54.75	84.78	59
CvT	60.47	49.42	89.55	54.39	85.35	18
DAT-DenseNet	79.79	85.75	96.11	82.66	92.41	7

our method in this paper shows better results in the aortic dissection classification task.

Experiments to explore number of DCAM modules on model performance

Among the DAT module, the most important one is the DCAM. In the DCAM, the multi-head attention mechanism is introduced into the convolutional neural network. It achieves superior local feature extraction capability by this. Since in DAT, the DCAM module is allowed to be combined with the residual connection module to realize a cascade of multiple DCAM modules. So, in order to investigate the effect of the number of DCAMs on the classification task, we designed some comparison experiments. In the experiment, the number of DCAM modules was set to 1, 2, 3 and 4. The experiments were conducted using the model fusion method. In Table 4, we can see that when the number of DCAM layers increases, the performance of the model decreases instead. We analyzed that it may be due to the fact that there are too many Transformer layers in it resulting in extracting too many global features. Moreover, increasing the parameters in the model leads to the increase of training difficulty, which affects the local information feature extraction ability of CNN. Therefore, the number of DCAM in the DAT module is not suitable to be too more, otherwise it will affect the overall performance of the model.

Experiments to explore low-threshold zero-processing method

In order to investigate the effect of the low-threshold zero-processing method on the task of aortic dissection classification based on CTA images, We designed experiments to analyze the effect of datasets under different threshold treatments on the classification task. The thresholds ranged from 150 to 225 in interpolated steps of 25. In the experiments, we used model fusion method. The experiments are still performed using five-fold cross-validation. In Table 5, we can see that the classification effect is optimal at a threshold of 200. The CTA image visualization for different thresholding operation is shown in Fig. 5. From the analysis of the results, the aortic region cannot be highlighted in the CTA image

Table 4 Evaluation the number of DCAM for classification results of CTA images

Depth	REC (%)	PRE (%)	SPE (%)	F1 (%)	ACC (%)
1	79.79	85.75	96.11	82.66	92.41
2	76.63	80.90	94.69	78.71	90.60
3	69.28	83.83	95.67	75.86	89.22
4	71.10	72.33	92.02	71.71	87.28

Table 5 Evaluation of thresholding operation on CTA image classification results

Threshold	REC (%)	PRE (%)	SPE (%)	F1 (%)	ACC (%)
–	44.26	85.97	92.24	58.43	67.39
150	70.22	80.49	95.01	75.01	89.39
175	74.28	78.66	94.09	76.41	89.60
200	79.79	85.75	96.11	82.66	92.41
225	78.87	75.59	92.53	77.20	89.43

when the threshold is set too small. When the threshold is set too large, some information around the aortic region is lost, as well as increasing the risk of distinguishing the spine section as an aortic dissection. Therefore, we can say that for CTA images, appropriate threshold treatment for images will improve the model's ability to extract features and thus optimize the model classification results.

Discussion

During the investigation of automated diagnostic methods for aortic dissection, we found that the task of segmentation of aortic dissection occupies a large portion of the work. MOLS-Net [11] is the only work that performs multi-label segmentation of the true and false lumen, and it even performs a fine segmentation of the lung tissues in the abdominal cavity. In our communication with clinicians, we were informed that such fine segmentation is not necessary for rapid diagnosis, so we tried to use a simple thresholding method to replace this part of the work. Threshold processing can obtain 85 % accuracy in aortic segmentation task without considering true-false lumen multilabel segmentation. As mentioned earlier, the main goal of this study is to improve the efficiency of the model, thereby assisting physicians in increasing the speed of diagnosis, minimising model inference time, and mitigating computational costs. This goal has been significantly and efficiently achieved with the classification model parallel architecture. We have counted the inference speed and classification accuracy of the model for individual patients, and the results, as shown in the Table 6, also demonstrate that parallel architectures are indeed effective in achieving accurate patient diagnosis using shorter durations.

In the past few years, various attention modules have been proposed to enhance the accuracy of models. Attention mechanisms have evolved through channel attention such as SE [26] and ECA [27], and hybrid domain attention such as CBAM [28] and GAM [29], which are able to capture key information according to different needs. In 2017, Transformer used a Self-Attention based model architecture, which achieves even better parallel computing capabilities.

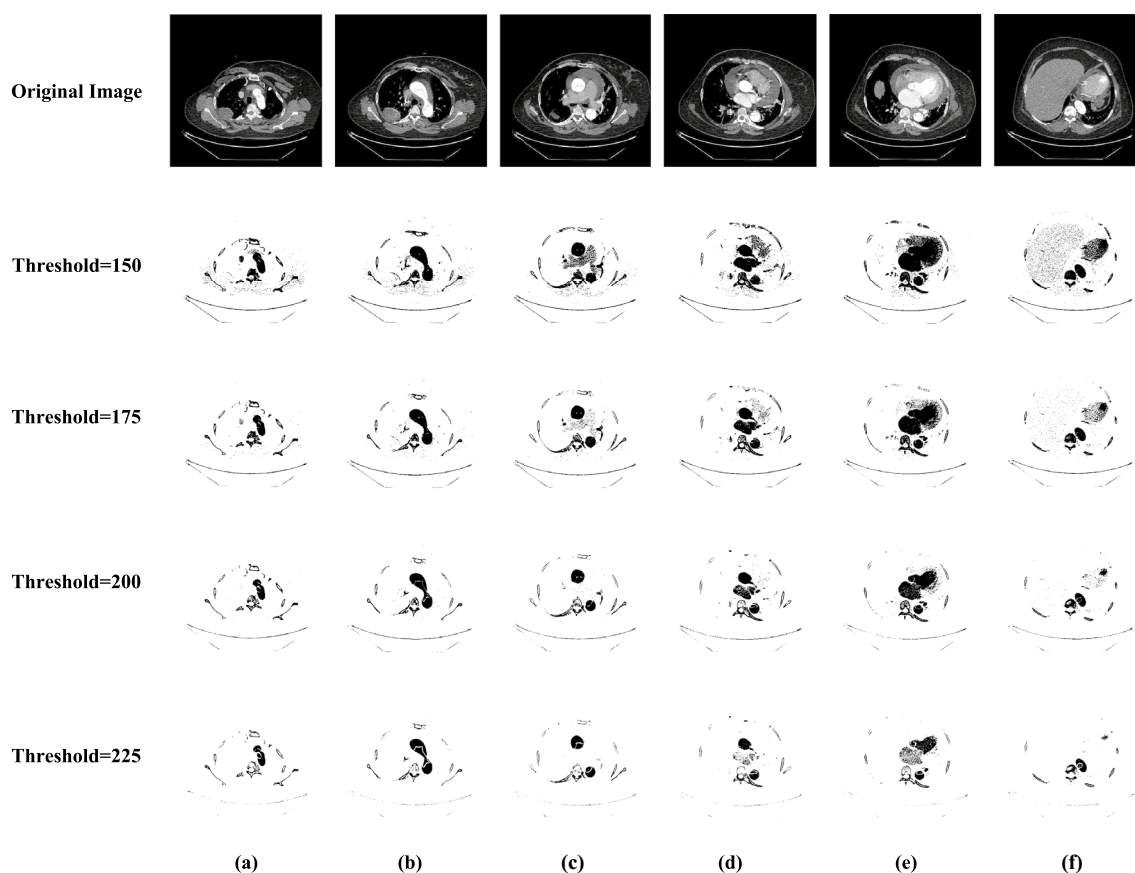


Fig. 5 Results of different thresholding operations for CTA images

Table 6 Statistics on the efficiency of different methods for diagnosing individual patients

Model	Inference time(s)	ACC (%)
Ding	5.65	90.74
Wei	4.22	90.86
Vatsala's	4.90	91.14
Akinori's	3.77	91.89
Our non-parallel architecture	6.43	90.50
Our parallel architecture	2.89	92.41

In the study of performance limitations in this paper, we consider that the linear mapping in the Transformer model architecture leads to a large number of model parameters. In addition, the large data requirements make it difficult to carry out research work on small datasets of medical images. We propose a lightweight attention module called DAT that combines Transformer ideas with convolutional neural networks. We compare the classification effects of applying different attention modules based on the DenseNet network. It should be noted that the DAT module involves the replacement of the Dense Block in the first layer, none of the other attention modules involve the replacement of DenseNet structures. From the analysis of the Table 7, DAT

Table 7 Evaluation of different attention modules on CTA image classification results

Model	REC (%)	PRE (%)	SPE (%)	F1 (%)	ACC (%)
DenseNet+SE	77.64	78.93	92.16	78.32	90.74
DenseNet+CBAM	77.37	78.17	91.72	78.39	90.86
DenseNet+ECA	80.25	82.05	92.21	80.75	91.14
DenseNet+GAM	79.50	85.92	94.95	82.04	91.89
DAT-DenseNet	79.79	85.75	96.11	82.66	92.41

has excellent performance compared to the other attention modules.

One of the most important aspects in the study of multi-model fusion methods is how to aggregate the results of multiple models into a single result for a patient. We compared the aortic dissection diagnostic algorithm proposed by Tomoki [20] and found that it uses the maximum number of consecutive images in a series of CTA images with an image level probability greater than or equal to 0.5 defined as the patient's outcome, and that it does not provide an explanation for the 0.5 cut-off threshold in its paper. Clinicians have also stated that this method of setting a fixed probability for image level is not well justified. In our study, the output features of the two classification models are firstly spliced firstly on a patient basis, and then deeply fused using the feature fusion module to achieve the purpose of refining the features, and finally arrive at the patient's diagnosis. This approach we interpreted as multiple image features of the patient mapping together to the patient results, and this interpretation was also recognised by the doctors in our partner hospitals, and we also confirmed the superiority of this aggregation approach in our experiments.

However, the dataset used in the experiment is small in size, and we trained on only 80 patients. Whether the methodology of the present study guarantees diagnostic effectiveness in data sets from a wider range of sources is worthy of further investigation. In the clinical setting, the amount of contrast agent and the model of imaging equipment used during CTA examinations can have an impact on CTA images, especially after using thresholding methods. We may be able to address the diagnostic efficiency of this study's method in large-scale datasets in the future through the collection of multi-centre data and image enhancement standardisation. In addition, if the patient suffers from other cardiovascular diseases, this may also have an impact on the automatic diagnosis of aortic dissection. These are also ignored by us in this study. Expanding the scope of patient inclusion might help with this limitation. Of course, this requires further validation.

Conclusion

In this paper, we propose an algorithm based on CTA images to distinguish whether the patient has aortic dissection. The proposed algorithm has the following advantages: The DAT-DenseNet model in this paper incorporates a convolutional neural network and multi-head attention mechanism of Transformer. It reduces the model's parameters and achieves excellent feature extraction. We innovatively arranged two classification models in parallel to achieve accurate classification of images without the need for aortic segmentation. Finally, through the feature fusion module, the

image features of the two classification models are spliced and fused on a patient-by-patient basis and aggregated into patient diagnosis results. In the experimental section, We designed the comparative experiment of classification models, the comparative experiment of model fusion methods, exploratory experiment for the number of DCAM and the exploratory experiment for low-threshold zero-processing methods. Ultimately, we confirmed the excellent performance of the DAT-DenseNet model and the model fusion method in the aortic dissection diagnosis task through the above experiments.

Funding The study was supported by LiaoNing Revitalization Talents Program (XLYC2008004) and the Shenyang Science and Technology Project (20-205-4-001).

Data availability The image data from the experiment are not open for public access due to the requirements of our partner hospital

Declarations

Conflict of interest All authors disclosed no relevant relationships.

Ethical approval The study received ethical approval from Medical Ethics Committee of the General Hospital of the Northern Theater of Operations of the Chinese People's Liberation Army (Approval No.: Y[2023]No.(160)).

References

1. DeSanctis RW, Doroghazi RM, Austen WG, Buckley MJ (1987) Aortic dissection. *N Engl J Med* 317(17):1060–1067
2. Golledge J, Eagle KA (2008) Acute aortic dissection. *Lancet* 372(9632):55–66
3. Juraszek A, Czerny M, Rylski B (2022) Update in aortic dissection. *Trends Cardiovasc Med* 32(7):456–461
4. Zhou SK, Greenspan H, Davatzikos C, Duncan JS, Van Ginneken B, Madabhushi A, Prince JL, Rueckert D, Summers RM (2021) A review of deep learning in medical imaging: Imaging traits, technology trends, case studies with progress highlights, and future promises. *Proc IEEE* 109(5):820–838
5. Lee N, Tek H, Laine AF (2008) True-false lumen segmentation of aortic dissection using multi-scale wavelet analysis and generative-discriminative model matching. In: *Medical Imaging 2008: Computer-aided diagnosis*, vol 6915, pp 878–888. SPIE
6. Gayhart M, Arisawa H et al (2013) Automated detection of healthy and diseased aortae from images obtained by contrast-enhanced CT scan. *Comput Math Methods Med* 2013:1
7. Dehghan E, Wang H, Syeda-Mahmood T (2017) Automatic detection of aortic dissection in contrast-enhanced CT. In: *2017 IEEE 14th international symposium on biomedical imaging (ISBI 2017)*, pp 557–560. IEEE
8. Yi Y, Mao L, Wang C, Guo Y, Luo X, Jia D, Lei Y, Pan J, Li J, Li S, et al. (2021) Early detection of aortic dissection on non-contrast CT: the combination of deep learning and morphologic characteristics
9. He K, Zhang X, Ren S, Sun J (2016) Deep residual learning for image recognition. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp 770–778

10. Xu X, He Z, Niu K, Zhang Y, Tang H, Tan L (2019) An automatic detection scheme of acute stanford type a aortic dissection based on dcnn in cta images. In: Proceedings of the 2019 4th international conference on multimedia systems and signal processing, pp 16–20
11. Zhou Q, Qin J, Xiang X, Tan Y, Ren Y (2022) Mols-net: Multi-organ and lesion segmentation network based on sequence feature pyramid and attention mechanism for aortic dissection diagnosis. *Knowl-Based Syst* 239:107853
12. Ye S, Zhijun F, Yongbin G (2020) Multi-model fusion of coronary cta segmentation based on attention mechanism. *J Front Comput Sci Technol* 14(9):1602
13. Long J, Shelhamer E, Darrell T (2015) Fully convolutional networks for semantic segmentation. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 3431–3440
14. Hata A, Yanagawa M, Yamagata K, Suzuki Y, Kido S, Kawata A, Doi S, Yoshida Y, Miyata T, Tsubamoto M et al (2021) Deep learning algorithm for detection of aortic dissection on non-contrast-enhanced ct. *Eur Radiol* 31:1151–1159
15. Ding Y, Tang X, Zhuang Y, Mu J, Chen S, Liu S, Feng S, Chen H (2023) Leukocyte subtype classification with multi-model fusion. *Med Biol Eng Comput* 1:1–12
16. Hearst MA, Dumais ST, Osuna E, Platt J, Scholkopf B (1998) Support vector machines. *IEEE Intell Syst Appl* 13(4):18–28
17. Anand V, Gupta S, Koundal D, Singh K (2023) Fusion of u-net and cnn model for segmentation and classification of skin lesion from dermoscopy images. *Expert Syst Appl* 213:119230
18. Wei M, Wu Q, Ji H, Wang J, Lyu T, Liu J, Zhao L (2023) A skin disease classification model based on densenet and convnext fusion. *Electronics* 12(2):438
19. Huang G, Liu Z, Van Der Maaten L, Weinberger KQ (2017) Densely connected convolutional networks. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 4700–4708
20. Wada T, Takahashi M, Matsunaga H, Kawai G, Kaneshima R, Machida M, Fujita N, Matsuoka Y (2023) An automated screening model for aortic emergencies using convolutional neural networks and cropped computed tomography angiography images of the aorta. *Int J Comput Assist Radiol Surg* 18(12):2253–2260
21. Simonyan K, Zisserman A (2014) Very deep convolutional networks for large-scale image recognition. arXiv preprint [arXiv:1409.1556](https://arxiv.org/abs/1409.1556)
22. Loshchilov I, Hutter F (2017) Decoupled weight decay regularization. arXiv preprint [arXiv:1711.05101](https://arxiv.org/abs/1711.05101)
23. Krizhevsky A, Sutskever I, Hinton GE (2012) Imagenet classification with deep convolutional neural networks. In: Advances in neural information processing systems 25
24. Liu Z, Lin Y, Cao Y, Hu H, Wei Y, Zhang Z, Lin S, Guo B (2021) Swin transformer: hierarchical vision transformer using shifted windows. In: Proceedings of the IEEE/CVF international conference on computer vision, pp 10012–10022
25. Wu H, Xiao B, Codella N, Liu M, Dai X, Yuan L, Zhang L (2021) Cvt: introducing convolutions to vision transformers. In: Proceedings of the IEEE/CVF international conference on computer vision, pp 22–31
26. Hu J, Shen L, Sun G (2018) Squeeze-and-excitation networks. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 7132–7141
27. Wang Q, Wu B, Zhu P, Li P, Zuo W, Hu Q (2020) Eca-net: efficient channel attention for deep convolutional neural networks. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 11534–11542
28. Woo S, Park J, Lee J-Y, Kweon IS (2018) Cbam: convolutional block attention module. In: Proceedings of the European Conference on Computer Vision (ECCV), pp 3–19
29. Liu Y, Shao Z, Hoffmann N (2021) Global attention mechanism: retain information to enhance channel-spatial interactions. arXiv preprint [arXiv:2112.05561](https://arxiv.org/abs/2112.05561)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.